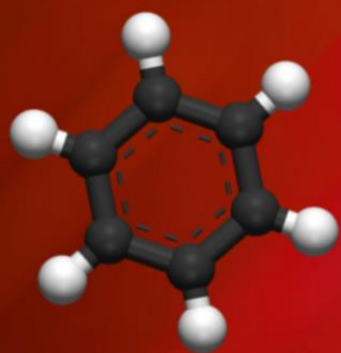




TurbOmics & TurboPutative

A web-server for multi-omics analysis, data handling and metabolite classification in
untargeted metabolomics

This website is free to use and open-source to all users and there is no login requirement

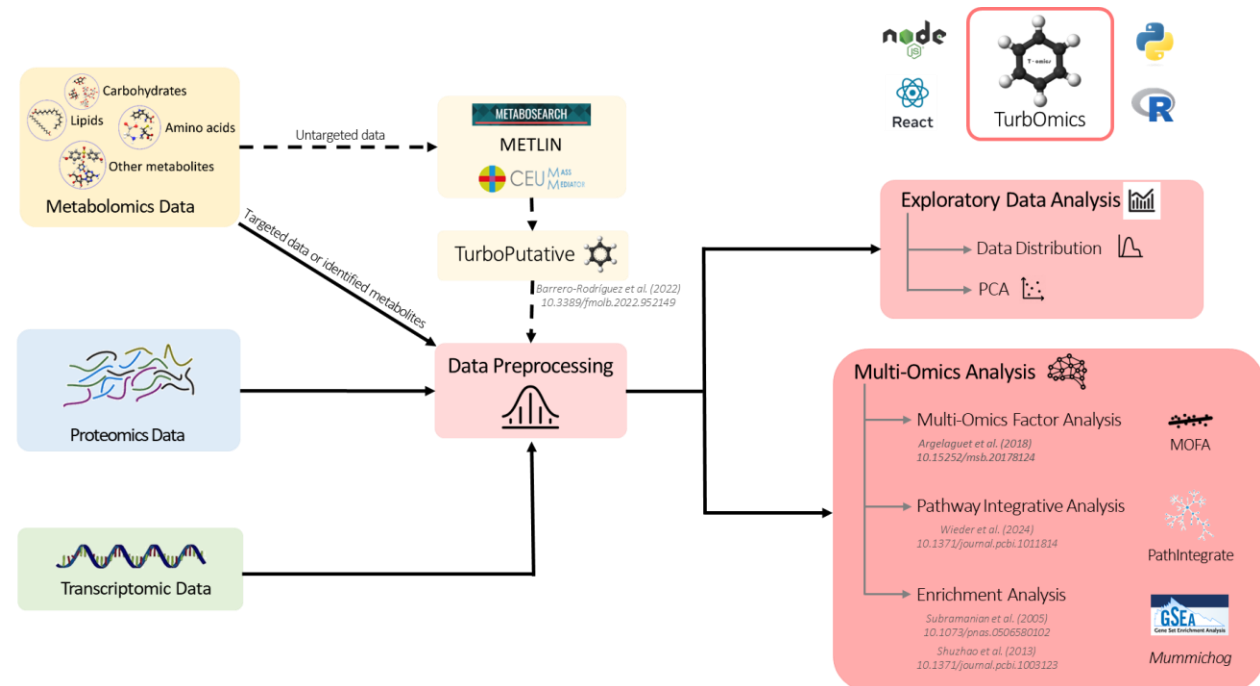
A step-by-step Tutorial

Overview

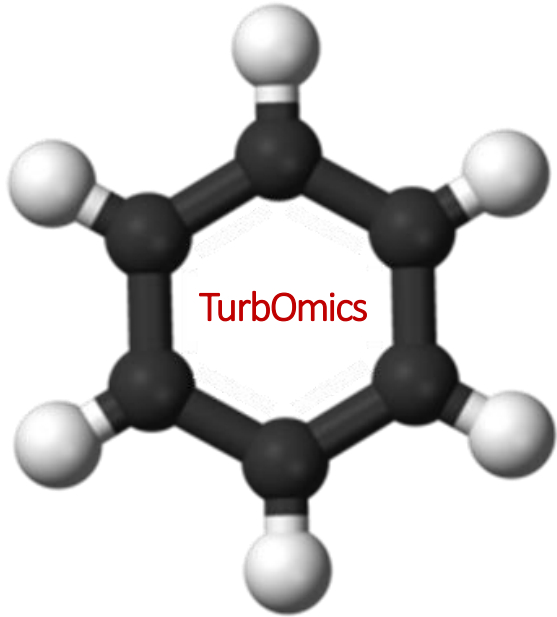
TurbOmics is an intuitive web-based platform designed to facilitate the multi-omics analysis of metabolomics data alongside proteomics and transcriptomics. It supports both **identified metabolites** from targeted-based experiments (LC/CE-MS) as well as those where the metabolite's identification is available (i.e. NMR or GC-MS). Additionally, TurbOmics offers tools to **putatively annotate** features from untargeted metabolomics based on their mass-to-charge (m/z) ratio by sequentially and internally running CMM and TurboPutative.

The platform provides a streamlined workflow for **exploratory data analysis** and **multi-omics integration**, through three main modules:

- **Multi-Omics Factor Analysis**
- **Pathway Integrative Analysis**
- **Enrichment Analysis**



Content



❑ INTRODUCTION

❑ EXAMPLES

❑ Case study (1): ATHEROSCLEROSIS

UNTARGETED LIPIDOMICS, PROTEOMICS and TRANSCRIPTOMICS from lorenzo *et al*, Nature 2021

- DATA PREPARATION
- EXPLORATORY ANALYSIS
- MULTI-OMICS FACTOR ANALYSIS
- ENRICHMENT ANALYSIS

❑ Case study (2): COVID-19

TARGETED METABOLOMICS and PROTEOMICS from Su *et al.*, Cell 2020

- DATA PREPARATION
- PATHWAY INTEGRATIVE ANALYSIS

❑ SUMMARY

Introduction

Background

Multi-omics integration provides a **comprehensive view of biological systems**, but current bioinformatics tools struggle to incorporate metabolomics data, especially for untargeted annotation and for offering alternatives to predefined knowledge bases. Moreover, advanced integration methods often require programming and statistical expertise, limiting their accessibility to non-specialist users.

The web application

TurbOmics, is a user-friendly web-based platform that enables researchers with diverse backgrounds **to analyze transcriptomics, proteomics, and metabolomics/lipidomics data** using an **integrative workflow** that includes advanced algorithms for **multi-omics integration**, while addressing key challenges associated with untargeted metabolomics data.

Data Formats

TurbOmics uses a structured data frame system composed of three types of tables that should be provided **in TSV format** :

- i. **Experimental metadata** (e.g. treatment, sex, time);
- ii. **Omics metadata** (e.g. *ChEBI ID*, *gene names*, annotations, RT, categories)
- iii. **Omics quantification** (abundancies or concentrations).

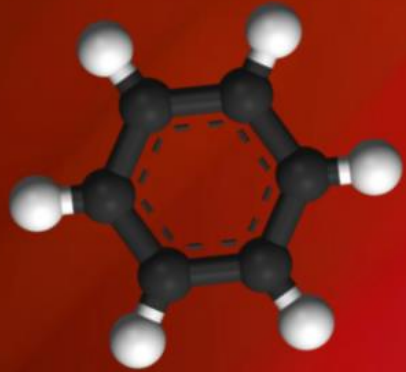
Expected Results

TurbOmics enables users to **map biomolecules** from different omics layer, **facilitating biological interpretation of multiomics results**. Users can explore and export data tables and visualize plots as results of:

- i. Exploratory Analysis
- ii. MOFA
- iii. Pathway Integrative Analysis
- iv. Enrichment Analysis

Try TurbOmics

GO to <https://proteomics.cnice.es/TurboPutative/home>



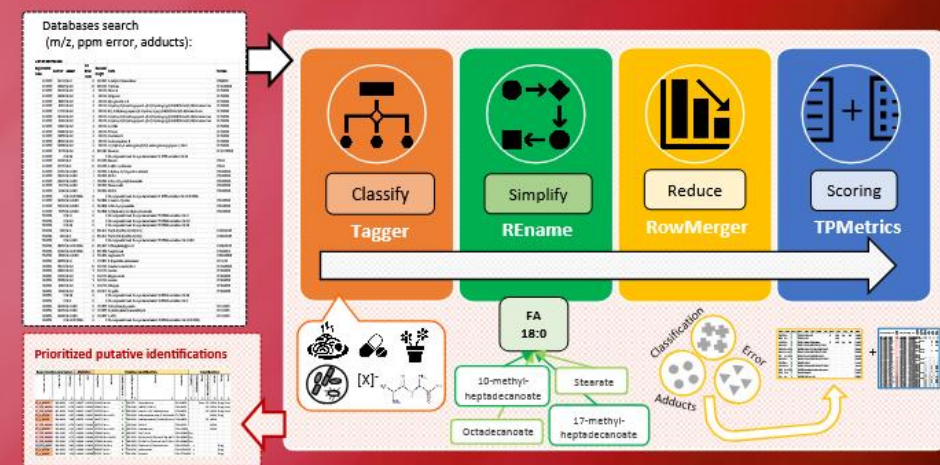
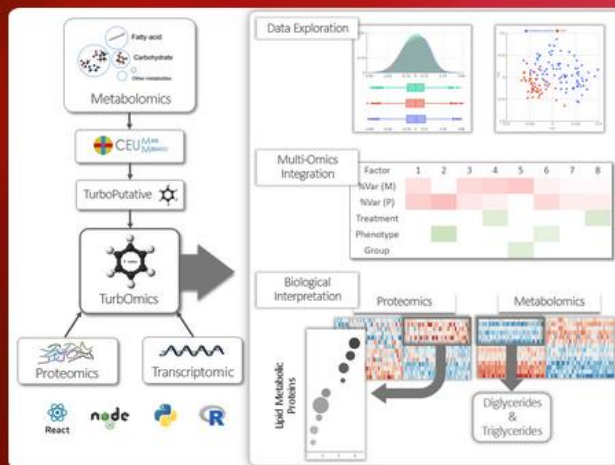
TurbOmics & TurboPutative

A web-server for multi-omics analysis, data handling and metabolite classification in untargeted metabolomics

This website is free to use and open-source to all users and there is no login requirement

 Try TurbOmics

 Try TurboPutative



Application to multi-omics datasets including untargeted lipidomics data

nature

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Article | Published: 02 December 2020

ALDH4A1 is an atherosclerosis auto-antigen targeted by protective antibodies

[Cristina Lorenzo](#), [Pilar Delgado](#), [Christian E. Busse](#), [Alejandro Sanz-Bravo](#), [Inmaculada Martos-Folgado](#), [Elena Bonzon-Kulichenko](#), [Alessia Ferrarini](#), [Ileana B. Gonzalez-Valdes](#), [Sonia M. Mur](#), [Raquel Roldán-Montero](#), [Diego Martinez-Lopez](#), [Jose L. Martin-Ventura](#), [Jesús Vázquez](#), [Hedda Wardemann](#) & [Almudena R. Ramiro](#) 

[Nature](#) **589**, 287–292 (2021) | [Cite this article](#)

19k Accesses | **124** Citations | **128** Altmetric | [Metrics](#)

- UNTARGETED LIPIDOMICS
- PROTEOMICS
- TRANSCRIPTOMICS

i. New data upload

To get started, users can upload their experimental data directly from the **Main Page**. Previous analyses can be accessed using the **Find Job** field by entering the corresponding job ID. Once a dataset is loaded, the **Results** section provides access to a suite of tools for in-depth analysis and interpretation.

HomeTurboPutativeTurbOmicsWeb ServicesHelpContact us

Search Job

Create new job

(1) Data files upload

Load the Untarget Lipidomic sample data

Load the Target Metabolomic sample data

Upload from R

(2) Choose appropriate organism

Organism
Mouse - Mus musculus

Upload the experimental metadata (e.g., treatment, sex, time).
First column must contain sample IDs

Experimental Metadata
Upload or drop a file right here TSV
File name: experimental_metadata.tsv

Upload the metadata from each omics dataset (e.g., ChEBI ID, RT, UniProt ID, gene name).
Rows = biomolecules
Columns = information

Transcriptomic Metadata
Upload or drop a file right here TSV
File name: transcriptomic_metadata.tsv

Metabolomic/Lipidomic Metadata
Upload or drop a file right here TSV
File name: metabolomic_metadata.tsv

Proteomic Metadata
Upload or drop a file right here TSV
File name: proteomic_metadata.tsv

Upload the quantifications from each omics dataset (e.g., abundancies, concentrations).
Rows = biomolecules; Columns = samples IDs

Transcriptomic Quantifications
Upload or drop a file right here TSV
File name: transcriptomic_quantifications.tsv

Metabolomic/Lipidomic Quantifications
Upload or drop a file right here TSV
File name: metabolomic_quantifications.tsv

Proteomic Quantifications
Upload or drop a file right here TSV
File name: proteomic_quantifications.tsv

	n
Meta-variables	18
Transcriptomic features	11999 / 11999
Metabolomic features	465 / 465
Proteomic features	4114 / 4114

ii. Load example dataset

To get started, users can upload their experimental data directly from the **Main Page**. Previous analyses can be accessed using the **Find Job** field by entering the corresponding job ID. Once a dataset is loaded, the **Results** section provides access to a suite of tools for in-depth analysis and interpretation.

The screenshot shows the TurbOmics web interface with several annotations:

- (1a)** A green dashed box highlights the 'Load directly the example containing the metadata and the quantitative data from all omics layers' section, which includes links to 'Load the Untarget Lipidomic sample data' and 'Load the Target Metabolomic sample data'.
- (1b)** A blue box highlights the 'Download the sample data files' section, which includes a link to 'Upload from R'.
- (1c)** A red box highlights the 'Job ID: UntargetedSamplePaper' field in the top navigation bar.
- (2)** A yellow dashed box highlights the 'Organism' dropdown menu, which is set to 'Mouse - Mus musculus'.

The interface also features a 'Search Job' field and a 'Create new job' button. Below the navigation bar, there are six panels for uploading data:

- Experimental Metadata**: Upload or drop a file right here (TSV). File name: experimental_metadata.tsv
- Transcriptomic Metadata**: Upload or drop a file right here (TSV). File name: transcriptomic_metadata.tsv
- Metabolomic/Lipidomic Metadata**: Upload or drop a file right here (TSV). File name: metabolomic_metadata.tsv
- Proteomic Metadata**: Upload or drop a file right here (TSV). File name: proteomic_metadata.tsv
- Transcriptomic Quantifications**: Upload or drop a file right here (TSV). File name: transcriptomic_quantifications.tsv
- Metabolomic/Lipidomic Quantifications**: Upload or drop a file right here (TSV). File name: metabolomic_quantifications.tsv
- Proteomic Quantifications**: Upload or drop a file right here (TSV). File name: proteomic_quantifications.tsv

	n
Meta-variables	18
Transcriptomic features	11999 / 11999
Metabolomic features	465 / 465
Proteomic features	4114 / 4114

Example of uploaded untargeted lipidomics dataset by TSV format

HomeTurboPutativeTurbOmicsWeb ServicesHelpContact us

Search Job

Mouse_IDTreatmentBatch

LD2169A12Batch1

LD2170A12Batch1

LD2171A12Batch1

LD2243A12Batch2

LD2244A12Batch2

LD2245A12Batch2

LD2162B1-8Batch1

LD2163B1-8Batch1

Organism

Mouse - Mus musculus

Create new job

Experimental Metadata

Upload or drop a file right hereTSV

File name: experimental_metadata.tsv

Metabolomic/Lipidomic Metadata

Upload or drop a file right hereTSV

File name: metabolomic_metadata.tsv

Proteomic Metadata

Upload or drop a file right hereTSV

File name: proteomic_metadata.tsv

Metabolomic/Lipidomic Quantifications

Upload or drop a file right hereTSV

File name: metabolomic_quantifications.tsv

Proteomic Quantifications

Upload or drop a file right hereTSV

File name: proteomic_quantifications.tsv

n18

Meta-variables4

Transcriptomic features11999 / 11999

Metabolomic features2935 / 2935

Proteomic features4114 / 4114

fidApex m/zRT [min]PlatformMode

P1760.5846519.341C18POS

P2876.8008823.952C18POS

P3782.5688218.116C18POS

P4758.5689718.518C18POS

P6806.5686817.913C18POS

P7874.7848123.42C18POS

P8848.7696323.348C18POS

P9784.5839319.02C18POS

P10810.5993419.6C18POS

File name: transcriptomic_metadata.tsv

IDLD2169LD2170LD2171LD2243LD2244LD2245

P112.65812.87712.69412.95212.78913.034

P211.45311.12810.89110.86111.40310.520

P311.80211.96211.79011.77411.24811.823

P411.66011.88911.82011.95911.74111.795

P611.46311.85411.71911.64611.44211.686

P710.6019.94610.34410.22510.7619.830

P810.4909.86010.14910.05010.2149.721

P911.14611.27011.37511.35911.40611.126

P1010.94011.12510.86710.81710.55110.836

TSV

Data format for the three uploaded tables

The consistency and linkage between these tables rely on matching identifiers. Therefore, it is essential that:

- The sample identifiers in the first column of the experimental metadata table match the column headers (i.e., the first row) of the omic quantification tables.
- The biomolecule identifiers in the first column of both the omics metadata and omics quantification tables match exactly.

Experimental Metadata

Sample_ID	Condition	Treatment	Batch
S001	Control	None	Batch1
S002	Control	None	Batch1
S003	Control	None	Batch2
S004	Control	None	Batch2
S005	Control	Drug_A	Batch1
S006	Control	Drug_A	Batch1
S007	Control	Drug_A	Batch2
S008	Control	Drug_A	Batch2
S009	Control	Drug_B	Batch1
S010	Control	Drug_B	Batch1
S011	Control	Drug_B	Batch2
S012	Control	Drug_B	Batch2
S013	Disease	None	Batch1
S014	Disease	None	Batch1
S015	Disease	None	Batch2
S016	Disease	None	Batch2
S017	Disease	Drug_A	Batch1
S018	Disease	Drug_A	Batch1
S019	Disease	Drug_A	Batch2
S020	Disease	Drug_A	Batch2
S021	Disease	Drug_B	Batch1
S022	Disease	Drug_B	Batch1
S023	Disease	Drug_B	Batch2
S024	Disease	Drug_B	Batch2

Metabolomics Metadata

Metabolite_ID	Name	Biol_Function	Chemical_Class	ChEBI_ID	Pathway
HMDB0000122	Glucose	Energy metabolism	Carbohydrate	17234	Glycolysis
HMDB0000190	Lactate	Anaerobic metabolism	Organic acid	24996	Cori cycle
HMDB0000538	ATP	Cellular energy	Nucleotide	15422	Oxidative Phospho.
⋮	⋮			⋮	⋮

Metabolomics Quantification

Metabolite_ID	S001	S002	S003	...	S010	S011	S012
HMDB0000122	5.3	6.1	5.8	...	5.9	5.4	4.8
HMDB0000190	10.5	11.2	9.9	...	10.9	10.3	9.5
HMDB0000538	7.9	8.4	7.5	...	7.8	7.1	6.8
⋮	⋮	⋮	⋮	...	⋮	⋮	⋮
HMDB0000259	25.7	26.3	24.9	...	25.4	24.1	23.3
HMDB0000562	6.8	7.2	6.5	...	6.7	6.3	5.8
HMDB0000243	3.2	3.8	3.1	...	3.5	3.2	2.8

Transcriptomics Metadata

Transcript_ID	Gene_Name	Biol_Function	Chromosome	Pathway_Association
ENST00000269305	TP53	Tumor suppression	17	p53 signaling
ENST00000352993	BRCA1	DNA repair	17	Homologous recomb.
ENST00000229239	GAPDH	Glycolysis	12	Metabolism
⋮	⋮			⋮

Transcriptomics Quantification

Transcript_ID	S001	S002	S003	...	S010	S011	S012
ENST00000269305	12.3	14.2	11.8	...	12.8	11.9	10.7
ENST00000352993	8.7	9.1	8.5	...	8.4	7.9	7.2
ENST00000229239	20.4	21.2	19.8	...	19.5	18.6	17.3
⋮	⋮	⋮	⋮	...	⋮	⋮	⋮
ENST00000324450	10.5	11.2	9.9	...	10.3	9.5	8.7
ENST00000377970	7.9	8.4	7.5	...	7.1	6.8	6
ENST00000361624	25.7	26.3	24.9	...	24.1	23.3	22

iii. Normalization, scaling and Missing values Imputation

- **Normalization techniques** are used to correct for systemic and technical variations that can obscure biological signals and lead to misleading conclusions.
- Different omics datasets often exhibit varying value ranges due to differences in technological platforms and the normalization methods applied during data generation. These disparities can be addressed **standardizing** omics data by setting the mean of each biomolecule to 0 and the standard deviation to 1.
- Once these parameters are set, the user can launch the analysis > create new job

(5) [Create new job](#)

Transcriptomic Quantifications

Upload or drop a file right here TSV

File name: transcriptomic_quantifications.tsv

Apply:

Normalization: None

☐ Center & Scale

No missing value detected

User can apply different type of normalization and scaling

None (3)

Log2

Median & Log2

VSN

Metabolomic/Lipidomic Quantifications

Upload or drop a file right here TSV

File name: metabolomic_quantifications.tsv

Apply:

Normalization: None

☒ Center & Scale

Proteomic Quantifications

Upload or drop a file right here TSV

File name: proteomic_quantifications.tsv

Apply:

Normalization: None

☐ Center & Scale

No missing value detected

- TurbOmics handles **missing values** independently for each omic type. This design allows users to apply different filtering and imputation methods tailored to the technical and methodological characteristics of each omics dataset, increasing the flexibility and robustness of the workflow.

(4a) **Metabolomic Features vs MV Threshold**

(4b) **Selected MV Threshold: 0.2**

Imputation Type: KNN

Imputation Techniques: KNN, Minimum, Mean, Median

Choose from several imputation techniques to handle missing values in the remaining biomolecules

Users can interactively adjust this threshold and observe in real time how many biomolecules are retained.

11

Putative Annotation

TurbOmics provides two alternative approaches for handling putative annotations of metabolite:

a) User-provided annotations: Users can upload a **metabolomics metadata table** that includes identifications or putative annotations previously obtained using tools such as [MetaboSearch](#) or Ceu Mass Mediator ([CMM](#)), optionally simplified with TurboPutative (TPMetrics table). Option recommended for larger datasets).

b) Automated annotation generation: TurbOmics can automatically generate putative annotations for all detected features in a metabolomics experiment by internally executing [CMM](#) and [TurboPutative](#). Option recommended for dataset with less that 500 features.

Job ID: 45329iGzy4

Back to results

Create new job

n	18
Meta-variables	4
Transcriptomic features	11999 / 11999
Metabolomic features	465 / 465
Proteomic features	4114 / 4114

Transcriptomic Metadata

Upload or drop a file right here TSV

File name: transcriptomic_metadata.tsv

Obtain Putative Annotations from CMM & TurboPutative?

Metabolomic features will be annotated using [Ceu Mass Mediator](#) and simplified using [TurboPutative](#).

NO

YES

Proteomic Metadata

Upload or drop a file right here TSV

File name: proteomic_metadata.tsv

Transcriptomic Quantifications

Upload or drop a file right here TSV

File name: transcriptomic_quantifications.tsv

Metabolomic/Lipidomic Quantifications

Upload or drop a file right here TSV

File name: metabolomic_quantifications.tsv

Proteomic Quantifications

Upload or drop a file right here TSV

File name: proteomic_quantifications.tsv

Apply:

Normalization

None

Apply:

Normalization

None

Apply:

Normalization

None

Putative Annotation

To enable automated annotation, users must upload a metadata table containing the *m/z*, retention time (RT), and ionization mode (positive or negative) for each feature.

CMM compares the *m/z* values of detected features against in-house libraries and external databases such as [HMDB](#), [LipidMaps](#), [METLIN](#), and [KEGG](#). The resulting candidate annotations are then processed by the four modules of TurboPutative: **Tagger**, **REname**, **RowMerger**, and **TPMetrics**. The final output, **TPFilter**, provides the most likely annotation for each feature and is integrated with the user-supplied metabolomics/lipidomics metadata table.

i. Upload the metabolomics/lipidomics metadata table

Metabolomic Putative Annotations

(1)

Set Columns

m/z Column
Apex m/z

Retention Time Column
RT [min]

Ionisation Mode Column
Mode

Positive Ion Value
POS

Negative Ion Value
NEG

User data matrix

fid	Apex m/z	RT [min]	Platform	Mode
P1	760.58465	19.341	C18	POS
P1002	577.51885	18.018	C18	POS
P1004	784.58414	23.702	C18	POS
P1006	853.68138	21.396	C18	POS
P101	812.65201	21.395	C18	POS
P1012	850.57197	21.487	C18	POS

Metabolomic/Lipidomic Metadata

Annotate
Features >>

Choose the columns label in your data that contain *m/z*, retention time (RT), and ionization mode (positive or negative) for each feature

PUTATIVE ANNOTATIONS PARAMETERS (CMM)

(2)

Tolerance (ppm)

Number
5

Specify the parts-per-million (ppm) tolerances for *m/z*-based searches in CMM

Putative Annotation

ii. Select potential adducts for both positive and negative ionization modes

Positive Adducts

☐ Select All

(3)

Click on the buttons to choose the potential adducts that can be formed in the analytical set-up (.i.e. mobile phases) used by the researcher

Negative Adducts

☐ Select all

M+H M+2H M+Na M+K M+NH₄ M+H-H₂O M+H-NH₄ 2M+H 2M+Na M+H-HCOONa 2M+H-H₂O M+3H M+2H+Na
M+H+2K M+H+2Na M+3Na M+H+Na M+H+K M+ACN+2H M+2Na M+2ACN+2H M+3ACN+2H M+CH₃OH+H M+ACN+H
M+2Na-H M+IsoProp+H M+ACN+Na M+2K-H M+DMSO+H M+2ACN+H M+IsoProp+Na+H 2M+NH₄ 2M+K 2M+ACN+H
2M+ACN+Na 3M+H 3M+Na M+H-2H₂O M+NH₄-H₂O M+Li 2M+2H+3H₂O M+H+CH₃COOH M+H+CH₃COONa M+F+H

M-H M+Cl M+HCOOH-H M-H-H₂O M-H-HCOONa
M-H+CH₃COONa 2M-H M-3H M-2H M+Na-2H
M+K-2H M+CH₃COOH-H M+Br M+TFA-H
2M+HCOOH-H 2M+CH₃COOH-H 3M-H M+F M+F+H

Putative Annotations

iii. Select TPMetric parameters (Turboputative)

(6)

Annotate
Features >>

RT is used by the TPMetrics module to prioritize annotations based on correlation analyses between features.

These criteria are based on the concept that signals coming from the same molecular entity and those sharing the same biochemical class show a common elution behavior in the LC-MS and a similar correlation pattern across the samples.

➤ Once these parameters are set, the user can launch the analysis > annotate features

TURBOPUTATIVE PARAMETERS (TPMetrics)

Mode Positive

Metabolite Retention Time Window
(4a)
Number
0.1

Lipid Retention Time Window
(4b)
Number
2

Select expected RT window of features generated from the same compound and ionized with **different adduct**

Select the expected RT window of features belonging to the **same compound class** (e.g. LPC, CAR)

Metabolite Retention Time Window
(4a)
Number
0.1

Lipid Retention Time Window
(4b)
Number
2

Select expected RT window of features generated from the same compound and ionized with **different adduct**

Select the expected RT window of features belonging to the **same compound class** (e.g. LPC, CAR)

Specify the most likely adducts formed in MS for each compound class by entering the **adduct type** in the corresponding field.

Lipid Class and Possible Adducts
(5)

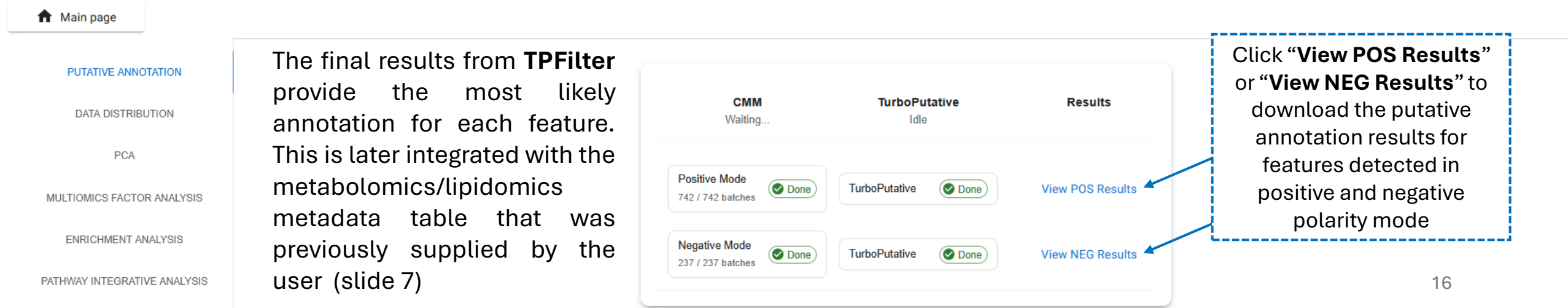
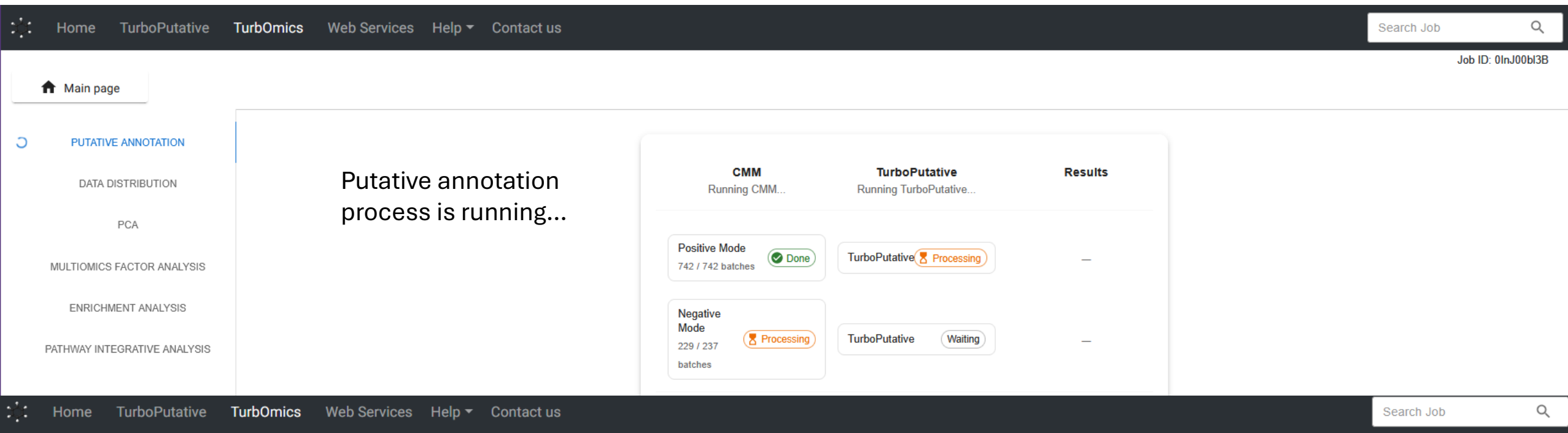
	Lipid	Adduct
1	PC	M+H
2	PC	M+Na
3	PC	M+K
4	PE	M+H
5	PE	M+Na
6	PG	M+NH4
7	PI	M+NH4
8	PA	M+NH4
9	LPC	M+H
10	LPC	M+Na
11	LPC	M+H-H2O
12	LPC	M+K
13	LPE	M+H
14	LPE	M+H-H2O
15	LPE	M+Na
16	SM	M+H

This option is only available for lipids since they can easily be classified using the abbreviations extracted from the Lipid Maps Structure Database (LMSD) (Sud et al., 2007) and the Goslin package (Kopczynski et al., 2020) incorporated into TurboPutative.

Lipid Class and Possible Adducts
(5)

	Lipid	Adduct
1	PC	M+HCOOH-H
2	PC	M+Cl
3	PE	M-H
4	PE	M-H+HCOONa
5	PS	M-H
6	PG	M-H
7	PG	M-H+HCOONa
8	PI	M-H
9	PI	M-H+HCOONa
10	PA	M-H
11	PA	M-H+HCOONa
12	LPC	M+HCOOH-H
13	LPC	M+Cl
14	LPE	M-H
15	LPE	M-H+HCOONa
16	LPS	M-H

Putative Annotation



Exploratory data analysis: Data Distribution

The Exploratory Data Analysis module provides researchers with an **initial overview and visualization of the data uploaded to the platform**.

In TurbOmics, exploratory analysis is conducted independently for each omics dataset.

To support this process, the platform includes two dedicated modules: Data Distribution and Principal Component Analysis (PCA).

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Job ID: 4eYOz8s8nD

Main page

PUTATIVE ANNOTATION

DATA DISTRIBUTION

PCA

MULTIOMICS FACTOR ANALYSIS

ENRICHMENT ANALYSIS

PATHWAY INTEGRATIVE ANALYSIS

Transcriptomics

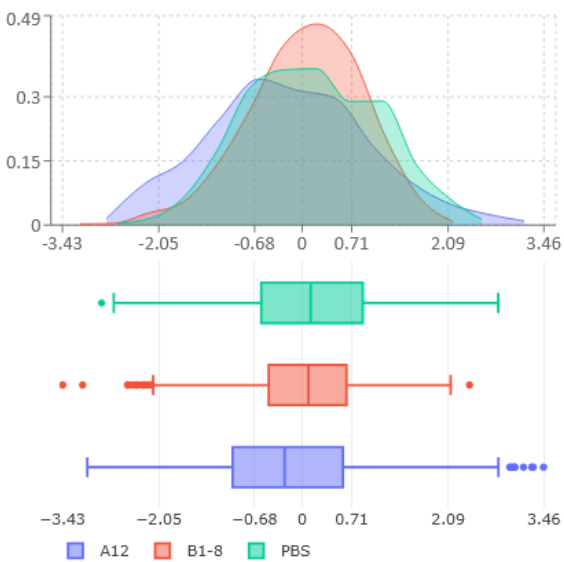
Metabolomics

Proteomics

Group by
Treatment

Download Normalized Data

Data Distribution



TP_Name

TG

	TP_Name
P2	TG 52:2 // TG 52:2; LMSD{ TG 52:2 }
P7	TG 52:3 // TG 52:3; LMSD{ TG 52:3 }
P8	TG 50:2 // TG 50:2; LMSD{ TG 50:2 }
P11	TG 54:3 // TG 54:3; LMSD{ TG 54:3 }
P15	TG 50:3 // TG 50:3; LMSD{ TG 50:3 }
P16	TG 50:1; LMSD{ TG 50:1 } // TG 50:1
P19	TG 52:4 // TG 52:4; LMSD{ TG 52:4 }
P20	TG 54:4 // TG 54:4; LMSD{ TG 54:4 }

(1) Select a **metadata category** to explore the distribution of a particular group of samples

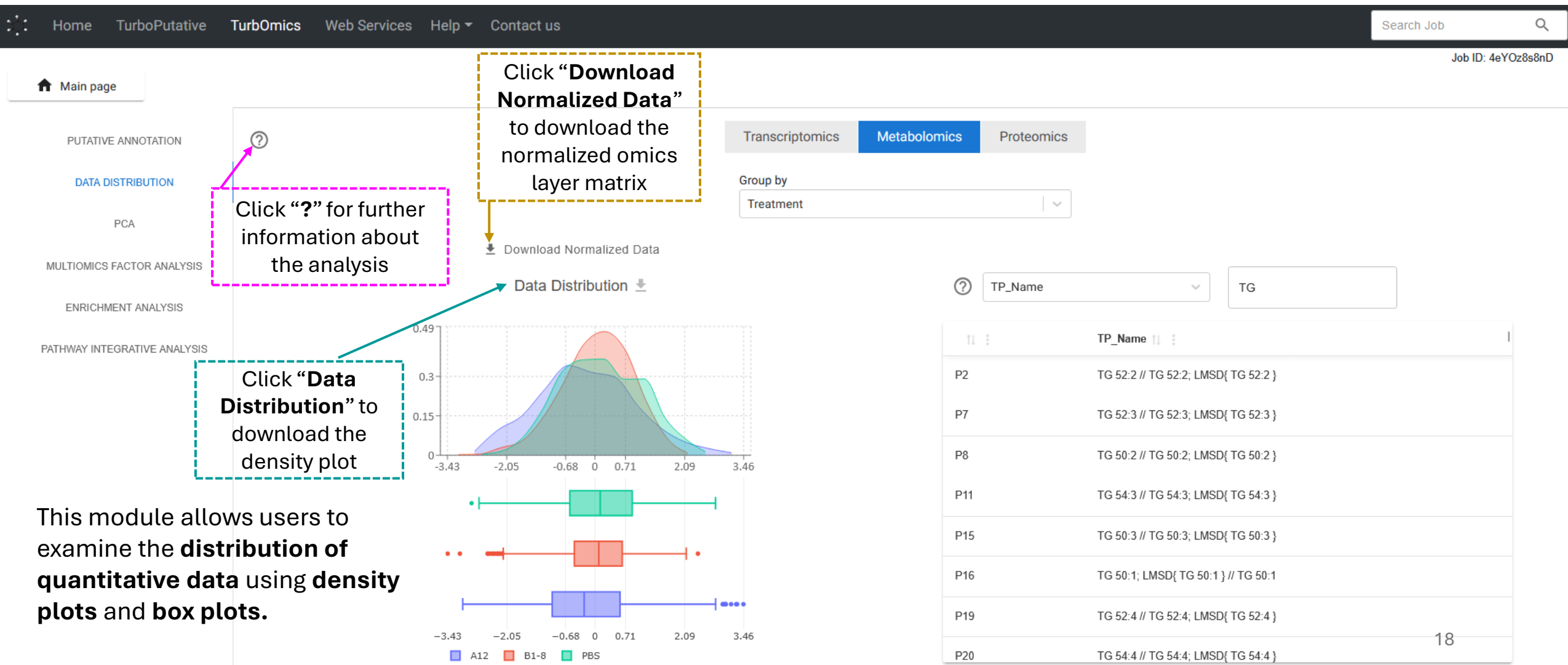
(2) Select the **column to be displayed** from the **Biomolecule Table lists** (e.g. putative annotation, ChEBI IDs, gene name, etc.)

(3) Users can filter a **set of biomolecules** based on a specific column from the Omics Metadata Table to displayed them in the density plot

This module allows users to examine the **distribution of quantitative data** using **density plots** and **box plots**.

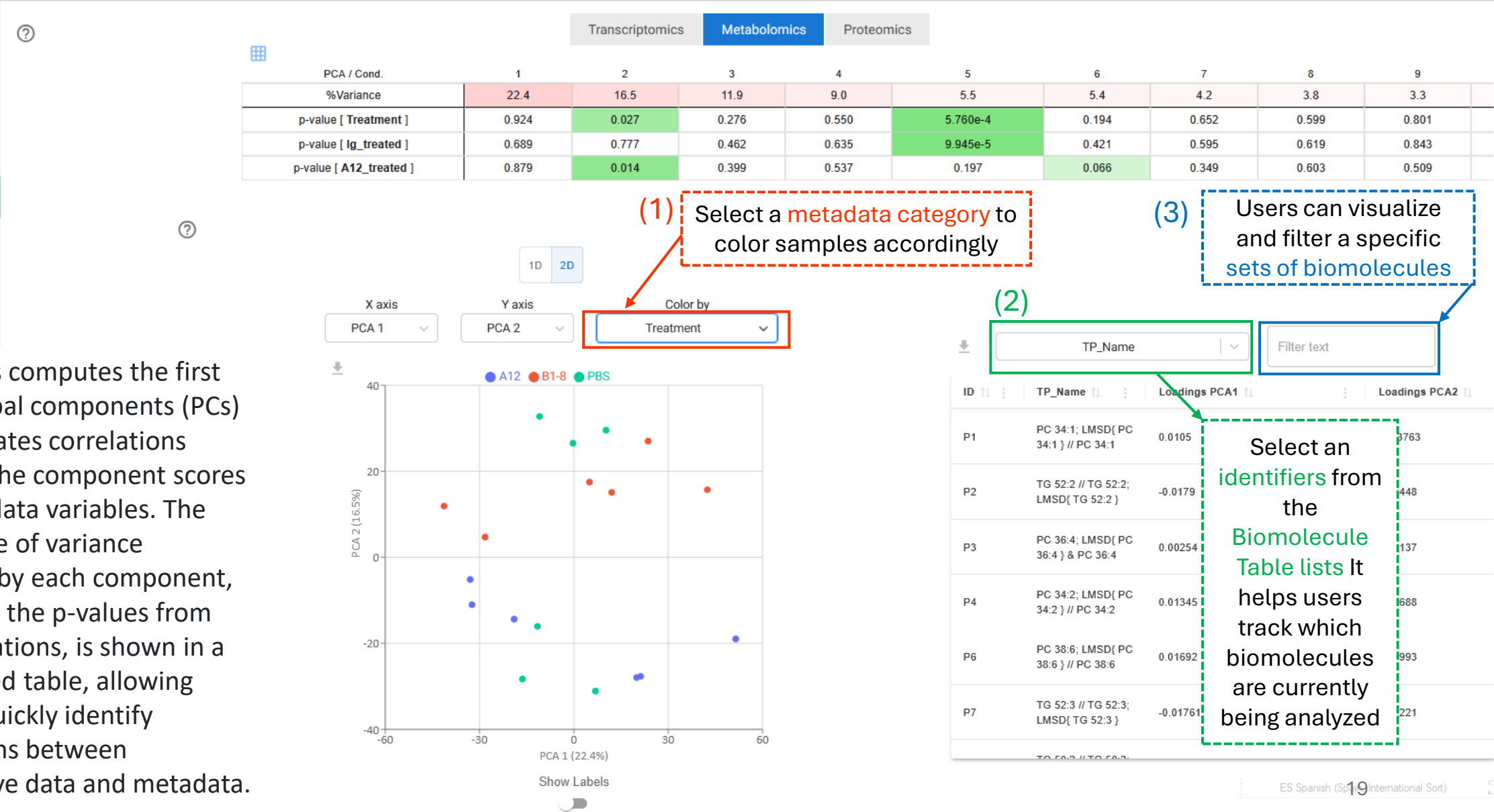
Exploratory data analysis: Data Distribution

The Exploratory Data Analysis module provides researchers with an **initial overview and visualization of the data uploaded to the platform**. In TurbOmics, exploratory analysis is conducted independently for each omics dataset. To support this process, the platform includes two dedicated modules: Data Distribution and Principal Component Analysis (PCA).



TP_Name
P2
P7
P8
P11
P15
P16
P19
P20

Exploratory data analysis: Principal Component Analysis (PCA)



TurbOmics computes the first ten principal components (PCs) and calculates correlations between the component scores and metadata variables. The percentage of variance explained by each component, along with the p-values from the correlations, is shown in a color-coded table, allowing users to quickly identify associations between quantitative data and metadata.

Exploratory data analysis: Principal Component Analysis (PCA)

Click “**Table icon**” to download the PCA table in CSV format



- PUTATIVE ANNOTATION
- DATA DISTRIBUTION
- PCA
- MULTIOMICS FACTOR ANALYSIS
- ENRICHMENT ANALYSIS
- PATHWAY INTEGRATIVE ANALYSIS

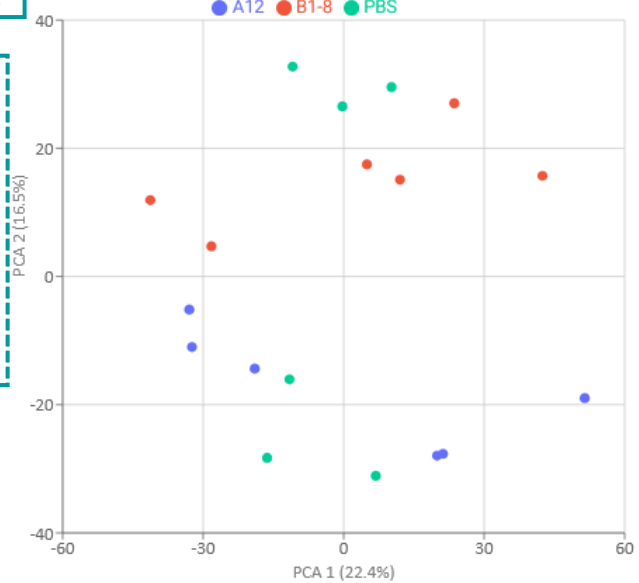


1D 2D

X axis PCA 1 Y axis PCA 2 Color by Treatment



Click on the “**download icon**” to download PCA scatter plot



Show Labels



Click on the “**download icon**” to download PCA loading table

ID	TP_Name	Loadings PCA1	Loadings PCA2
P1	PC 34:1; LMSD{ PC 34:1 } // PC 34:1	0.0105	-0.03763
P2	TG 52:2 // TG 52:2; LMSD{ TG 52:2 }	-0.0179	0.00448
P3	PC 36:4; LMSD{ PC 36:4 } & PC 36:4	0.00254	0.02137
P4	PC 34:2; LMSD{ PC 34:2 } // PC 34:2	0.01345	0.00688
P6	PC 38:6; LMSD{ PC 38:6 } // PC 38:6	0.01692	0.00993
P7	TG 52:3 // TG 52:3; LMSD{ TG 52:3 }	-0.01761	0.02221

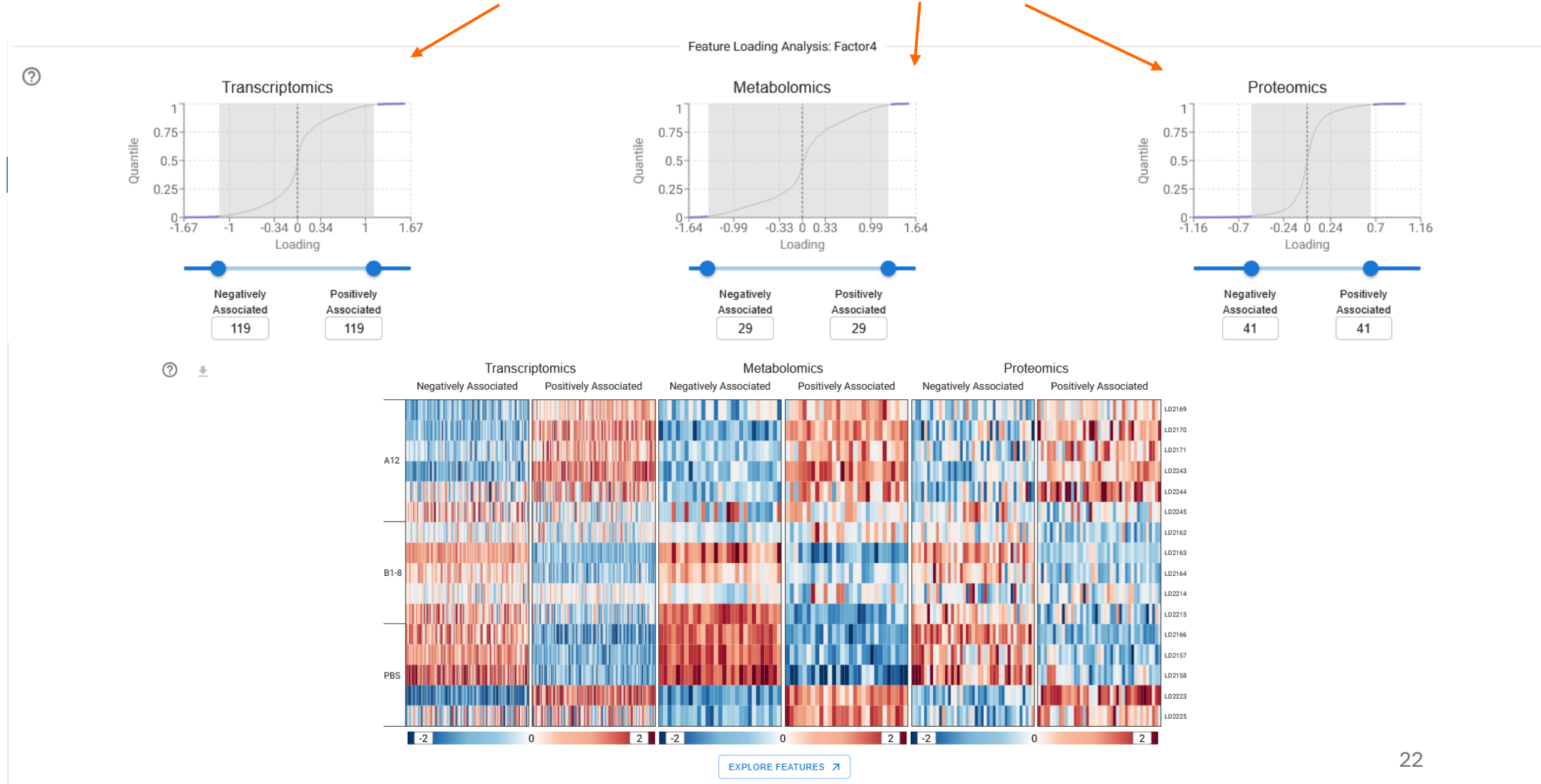
Multi-Omics Factor Analysis

The **Multi-Omics Factor Analysis** module enables the unbiased and integrative decomposition of multi-omics quantitative data into latent factors. This module is based on the [MOFA](#) framework, which identifies latent factors that capture the main sources of variation across the analyzed datasets. TurbOmics supports the visualization of sample subgroups and shared patterns across different omics layers.

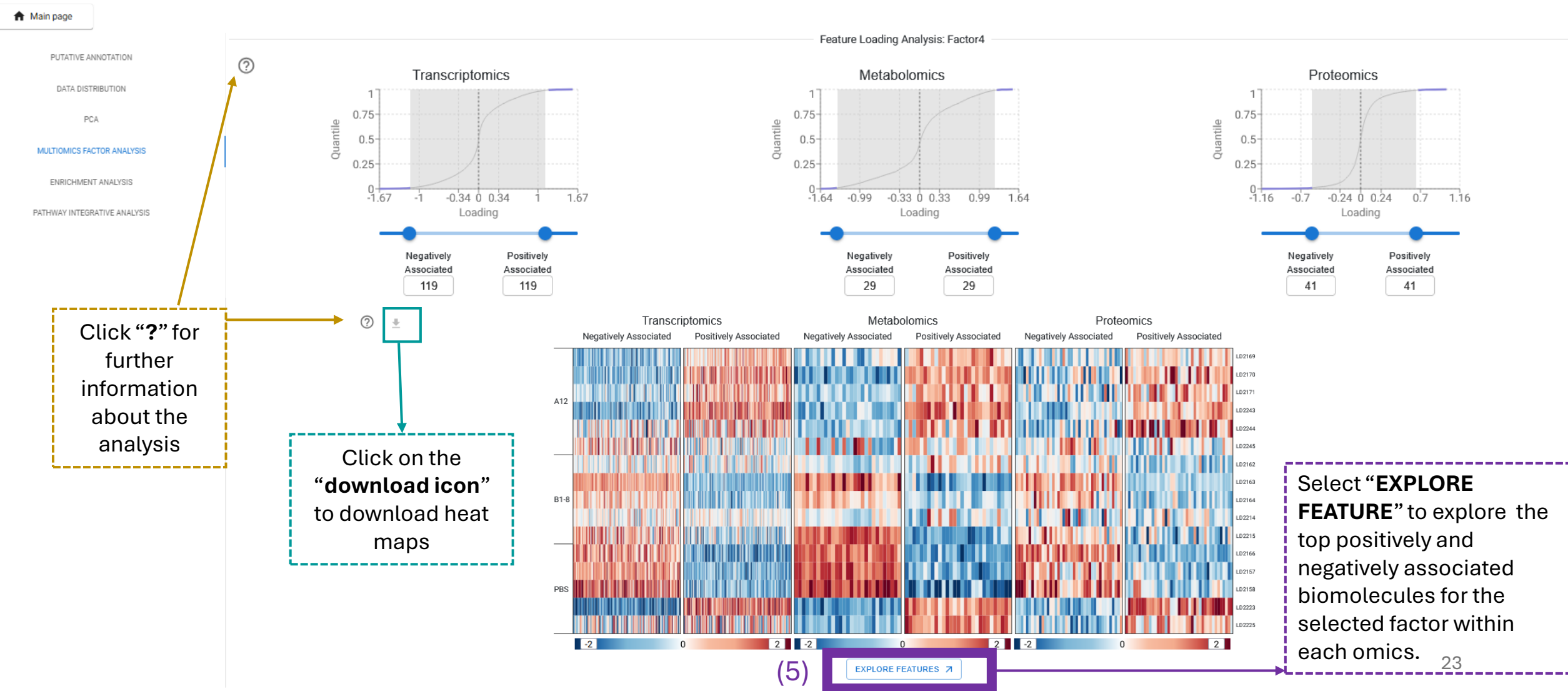


Multi-Omics Factor Analysis: Feature Loading Analysis

- (4) Users can examine the distribution of factor loadings for each omics dataset and select the top-n biomolecules with the highest positive or negative loadings for visualization in a heatmap.



Multi-Omics Factor Analysis: Feature Loading Analysis



Multi-Omics Factor Analysis

- Example of ORA (over-representation analysis) of the transcriptomics data within Factor 2 in MOFA

Explore Features: Factor2 vs Treatment

Transcriptomics Metabolomics Proteomics

Positively Associated Transcriptomics

Click “EXPORT TABLE” to download feature table

Users can filter the table introducing key words in the specific filter's tabs.

Click “bar icon” to hide/show table

Users can change the filter mode to visualize the data

Users can sort the data

Gene_id	Gene_symbol	Protein_name	Uniprot_ACC	Z(A12)	Z(B1-8)	Z(PBS)	Factor2
ENSMUSG00000036054	Sugp2	SURP and G-patch domain-containing prote	SUGP2_MOUSE	3.0117	2.5649	2.4398	2.4766
ENSMUSG00000035545	Leng8	Leukocyte receptor cluster member 8 hom	LENG8_MOUSE	5.7404	5.4333	5.4333	2.4289
ENSMUSG00000030131	Mug2	Murinoglobulin-2	MUG2_MOUSE		8.5354		2.326
ENSMUSG00000032637	Atxn2l	Ataxin-2-like protein	ATX2L_MOUSE		5.234	5.116	2.1715
ENSMUSG00000032637	Atxn2l	Leukotriene-B4 omega-hydroxylase 3	CP4FE_MOUSE		6.4147	7.0027	2.1026
ENSMUSG00000032637	Atxn2l	UDP-glucose:glycoprotein glucosyltransferase	UGGG1_MOUSE	5.707	5.1405	5.5348	2.0115
ENSMUSG00000032637	Atxn2l	AMP deaminase 2	AMPD2_MOUSE	4.8798	4.3953	4.6318	1.9805
ENSMUSG00000050541	Adra1b	Alpha-1B adrenergic receptor	ADA1B_MOUSE	5.5981	5.0734	5.3794	1.9734
ENSMUSG00000001424	Snd1	Staphylococcal nuclease domain-containing	SND1_MOUSE	6.9258	6.2178	6.8098	1.9732
ENSMUSG00000032198	Dock6	Dedicator of cytokinesis protein 6	DOCK6_MOUSE	4.2186	3.6034	3.9369	1.9152

Rows per page 10

This table lists the selected biomolecules and provides additional context by displaying the average quantitative value of each biomolecule across different sample groups.

Multi-Omics Factor Analysis

➤ Example of ORA (over-representation analysis) of the proteomics data within Factor 4 in MOFA

×

Explore Features: Factor4 vs Treatment

Transcriptomics

Metabolomics

Proteomics

?

GO Enrichment with g:Profiler

GO:MF

GO:BP

GO:CC

KEGG

REAC

microbody

peroxisome

Peroxisome

chaperonin-containing T-complex Association of

Select Column Containing ID

Select ID Format

Field

fid

Field

UNIPROT_GN_ACC

To aid in the biological interpretation of MOFA results, **ORA** is performed using databases such as KEGG, REACTOME, and GO. This analysis helps identify significantly enriched biological pathways, processes, or functions among the selected biomolecules.

EXPORT ALL DATA

ID	GN	Description
A2AKK5	Acnat1	acyl-coenzyme A amino acid N-acyltransferase 1
Q922Q1	Mtarc2	mitochondrial amidoxime reducing component 2
A2AKD6	Pxmp4	peroxisomal membrane protein 4
Q9JJW0	Pxmp4	peroxisomal membrane protein 4
Q99LC9	Pex6	peroxisomal biogenesis factor 6
Q8VCI5	Pex19	peroxisomal biogenesis factor 19
Q9DC50	Crot	camitine O-octanoyltransferase

EXPORT ALL DATA

Select GO

Filter by GO

GO:0042579

GO:CC

microbody

2.57e-8

GO:0005777

GO:CC

peroxisome

2.57e-8

KEGG:04146

KEGG

Peroxisome

6.35e-8

GO:0005832

GO:CC

chaperonin-containing T-complex

3.2e-5

REAC:R-MMU-390471

REAC

Association of TriC/CCT with target proteins during biosynthesis

4.17e-5

By clicking on a category, you can view all biomolecules associated it, with the ones selected for the factor highlighted in blue.

Multi-Omics Factor Analysis

➤ Example of ORA (over-representation analysis) of the proteomics data within Factor 4 in MOFA

×

Explore Features: Factor4 vs Treatment

Transcriptomics

Metabolomics

Proteomics

?

Click on the “download icon” to download the enrichment bar plot

GO Enrichment with g:Profiler

GO:MF

GO:BP

GO:CC

KEGG

REAC

microbody

peroxisome

Peroxisome

chaperonin-containing T-complex Association of

Click on the “download icon” to download the GO enrichment table

EXPORT ALL DATA

Select Column Containing ID

Select ID Format

Field

fid

Field

UNIPROT_GN_ACC

Click on the “download icon” to download category table

EXPORT ALL DATA

ID

GN

Description

acetyl-coenzyme A amino acid N-acyltransferase 1

mitochondrial amidoxime reducing component 2

peroxisomal membrane protein 4

peroxisomal membrane protein 4

peroxisomal biogenesis factor 6

peroxisomal biogenesis factor 19

carbamate O-octanoyltransferase

Select

GO

Filter by GO

Type

Filter by Type

Name

Filter by Name

FDR

0.1

GO:0042579

GO:CC

microbody

2.57e-8

GO:0005777

GO:CC

peroxisome

2.57e-8

KEGG:04146

KEGG

Peroxisome

6.35e-8

GO:0005832

GO:CC

chaperonin-containing T-complex

3.2e-5

REAC:R-MMU-390471

REAC

Association of Tric/CCT with target proteins during biosynthesis

4.17e-5

Enrichment Analysis

This module allows the user to perform Enrichment Analysis using the **Gene Set Enrichment Analysis (GSEA)** algorithm on each omics dataset independently. Users can switch between omics using the navigator bar at the top of the page to explore results for different data types. If you are working with **untargeted metabolomics/lipidomics data**, you can choose to perform enrichment analysis using the **Mummichog algorithm**.

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PATHWAY INTEGRATIVE ANALYSIS

Transcriptomics

Metabolomics

Proteomics

(1)

Users need to provide the following additional columns from the untargeted metabolomics/lipidomics metadata table: Apex m/z, Retention Time (min), Ion Mode

Mummichog

Select Apex m/z Column

Apex m/z

Select Retention Time Column (min)

RT column (min)

RT [min]

Select Ion Mode Column

Ion Mode Column

Mode

Positive Ion Value

Positive Ion

POS

Negative Ion Value

Negative Ion

POS

(2)

Users must specify the metric used for the enrichment analysis. Different options are available

Select Enrichment Ranking Metric

GSEA Ranking Metric

Mean difference

t-test

PCA

MOFA

Custom

Select Enrichment Ranking Metric

GSEA Ranking Metric

MOFA

Select MOFA Factor

MOFA Factor

Factor2

(3)

RUN QEA

Complement the analysis with your Pathways (Optional)

UPLOAD NEW PATHWAY

Uploaded: BA_mouse.gmt

Select Pathways

BA_human, BA_mouse

BA_human

BA_mouse

Optional

Users can upload custom pathways in GMT format, enabling the integration of personalized metabolite-based pathway definitions

27

FS Spanish (Spain, International Snt)

Enrichment Analysis: example using untargeted lipidomics data

Results of the Enrichment Analysis can be explored, providing insights into the biological pathways, processes, or functions that are significantly associated with the conditions or experimental groups in the study.

Click “**EXPORT ALL DATA**” to download the enrichment table

Enrichment Analysis: MOFA | Factor6

Custom KEGG REACTOME Mummichog (+) **Mummichog (-)**

[EXPORT ALL DATA](#)

☐	Pathway	pval	Overlap Size	Pathway Size	Features
	Filter by Pathway	Filter by ...	Filter by O...	Filter by P...	Filter by Features
	Filter Mode: Fuzzy	Filter Mode: Greater Than Or E	Filter Mode: Greater Than Or E	Filter Mode: Fuzzy	
☐	Arachidonic acid metabolism	6.72e-4	3	5	N238,N889
☐	Porphyrin metabolism	9.24e-3	1	1	N320
☐	Glycerophospholipid metabolism	1.50e-2	2	7	N238,N889
☐	Vitamin A (retinol) metabolism	1.61e-2	1	2	N889
☐	Omega-6 fatty acid metabolism	1.61e-2	1	2	N238,N889
☐	Leukotriene metabolism	2.36e-2	1	3	N238,N889
☐	Prostaglandin formation from arachidonate	3.04e-2	1	4	N238,N889
☐	De novo fatty acid biosynthesis	4.54e-2	1	12	N238,N889

Enrichment Analysis: example using proteomics data

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ENRICHMENT ANALYSIS

PATHWAY INTEGRATIVE ANALYSIS

Transcriptomics Metabolomics **Proteomics**

(1) Select column containing Proteomics ID

Proteomics ID column

UniProt_ID

Users need to specify the column in their proteomic metadata table that contains the biomolecule identifiers.

Select Enrichment Ranking Metric

GSEA Ranking Metric

Mean difference

t-test

PCA

MOFA

Custom

(2) Users must specify the metric used for the enrichment analysis.

Select GSEA Ranking Metric

GSEA Ranking Metric

MOFA

Select MOFA Factor

MOFA Factor

Factor4

(3) RUN QEA >

Enrichment Analysis: MOFA | Factor4

Custom HALLMARK GO:MF GO:CC GO:BP **KEGG** REACTOME

EXPORT ALL DATA

	Pathway	pval	padj	ES	NES	N. Leading Edge	Size
☐	Lysosome	2.39e-7	1.00e-4	-0.6794	-2.1791	22	57
☐	Peroxisome	6.83e-7	1.00e-4	0.6794	2.1046	27	62
☐	Propanoate metabolism	4.00e-4	3.75e-2	0.6996	1.8407	9	29
☐	Tight junction	6.00e-4	4.18e-2	-0.5444	-1.7862	29	63
☐	Proteasome	1.30e-3	7.15e-2	0.622	1.7728	18	40
☐	Protein processing in endoplasmic reticulum	1.40e-3	7.15e-2	0.501	1.6364	19	90
☐	Bacterial invasion of epithelial cells	1.90e-3	7.95e-2	-0.56	-1.6941	18	43
☐	Salmonella infection	2.10e-3	7.95e-2	-0.4155	-1.5004	25	120

Click "EXPORT ALL DATA" to download the enrichment table

✓ For Pathway Integrative Analysis see slides 36-40 of the present tutorial

Application to multi-omics datasets in a COVID-19 study, including targeted metabolomics data



Volume 183, Issue 6, 10 December 2020, Pages 1479-1495.e20

Article

Multi-Omics Resolves a Sharp Disease-State Shift between Mild and Moderate COVID-19

Yapeng Su¹, Daniel Chen¹, Dan Yuan^{1,2}, Christopher Lausted¹, Jongchan Choi¹,
Chengzhen L. Dai¹, Valentin Voillet^{3,4}, Venkata R. Duvvuri¹, Kelsey Scherler¹, Pamela Troisch¹,
Priyanka Baloni¹, Guangrong Qin¹, Brett Smith¹, Sergey A. Kornilov¹, Clifford Rostomily¹,
Alex Xu¹, Jing Li⁵, Shen Dong⁶, Alissa Rothchild⁷, Jing Zhou⁸...James R. Heath^{1,2,21}  

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<https://doi.org/10.1016/j.cell.2020.10.037> 

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 Open access

- TARGETED METABOLOMICS
- PROTEOMICS
- TRANSCRIPTOMICS

Load example dataset

To get started, users can upload their experimental data directly from the **Main Page**. Previous analyses can be accessed using the **Find Job** field by entering the corresponding job ID. Once a dataset is loaded, the **Results** section provides access to a suite of tools for in-depth analysis and interpretation.

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Search Job

Create new job

Load the Untarget Lipidomic sample data

Load the Target Metabolomic sample data

Upload from R

(1b) Download the sample data files

(1a) Load directly the example containing the metadata and the quantitative data from all omics layers

(2) Choose appropriate organism

Organism
Mouse - Mus musculus

Experimental Metadata

Upload or drop a file right here TSV

File name: experimental_metadata.tsv

n	18
Meta-variables	4
Transcriptomic features	11999 / 11999
Metabolomic features	465 / 465
Proteomic features	4114 / 4114

Transcriptomic Metadata

Upload or drop a file right here TSV

File name: transcriptomic_metadata.tsv

Metabolomic/Lipidomic Metadata

Upload or drop a file right here TSV

File name: metabolomic_metadata.tsv

Proteomic Metadata

Upload or drop a file right here TSV

File name: proteomic_metadata.tsv

Transcriptomic Quantifications

Upload or drop a file right here TSV

File name: transcriptomic_quantifications.tsv

Metabolomic/Lipidomic Quantifications

Upload or drop a file right here TSV

File name: metabolomic_quantifications.tsv

Proteomic Quantifications

Upload or drop a file right here TSV

File name: proteomic_quantifications.tsv

- ✓ For Normalization, Scaling and Missing Values Imputation see slide 11 of the present tutorial

Putative Annotation

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Back to results

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Organism

Human - Homo sapiens

Load the Untarget Lipidomic sample data

Load the Target Metabolomic sample data

Upload from R

Experimental Metadata

Metabolomic/Lipidomic Metadata

Proteomic Metadata

Transcriptomic Metadata

Transcriptomic Quantifications

Metabolomic/Lipidomic Quantifications

Proteomic Quantifications

Obtain Putative Annotations from CMM & TurboPutative?

Metabolomic features will be annotated using [CeU Mass Mediator](#) and simplified using [TurboPutative](#).

NO

YES

Apply:

Normalization

None

Center & Scale

Apply:

Normalization

None

Center & Scale

n

378

Meta-variables

6

Metabolomic features

750 / 1050

Proteomic features

454 / 464

In this example, metabolites identification are already provided

- ✓ For Exploratory analysis see pp.17-20 of the present tutorial
- ✓ For Multiomics Factor Analysis (MOFA) see pp. 21-26 of the present tutorial
- ✓ For Enrichment Analysis see pp. 27-29 of the present tutorial

Pathway Integrative Analysis

This section leverages the **PathIntegrate** library to identify pathways that are most strongly associated with a condition of interest. By using single-sample pathway analysis (ssPA) approaches, PathIntegrate transforms molecular-level abundance data into pathway-level matrices, enabling a more comprehensive understanding of how different pathways contribute to the observed biological outcomes. In this implementation, Principal Component Analysis (PCA) is used for summarizing the data at the pathway level.

➤ Single-View Framework

The screenshot displays the PathIntegrate web interface with the 'Single-View' analysis framework selected. The interface includes a navigation menu on the left with options like 'Main page', 'DATA DISTRIBUTION', 'PCA', 'MULTIOMICS FACTOR ANALYSIS', 'ENRICHMENT ANALYSIS', and 'PATHWAY INTEGRATIVE ANALYSIS'. The main content area is divided into three numbered steps:

- (1) 1. Select the Metadata**: A dashed orange box highlights this step. It contains a 'Metadata Column' dropdown set to 'Group', and two 'Group' dropdowns labeled 'Group 1' (set to 'He...') and 'Group 2' (set to 'C ...'). An annotation box above it says: 'Specify the **comparison groups** to be considered in the pathways analysis'.
- (2) 2. Select the Omics Mapping**: A dashed blue box highlights this step. It contains two dropdowns for 'Metabolomics ID Column' (set to 'ChEBI_ID') and 'Proteomics ID Column' (set to 'UniProt_ID'). Below these are 'ChEBI ID Type' and 'Uniprot Protein ID ID Type' dropdowns. An annotation box above it says: 'Select the **Identifiers** for each omics layer to use in the pathways analysis'.
- (3) RUN ANALYSIS >**: A purple box highlights the 'RUN ANALYSIS' button.

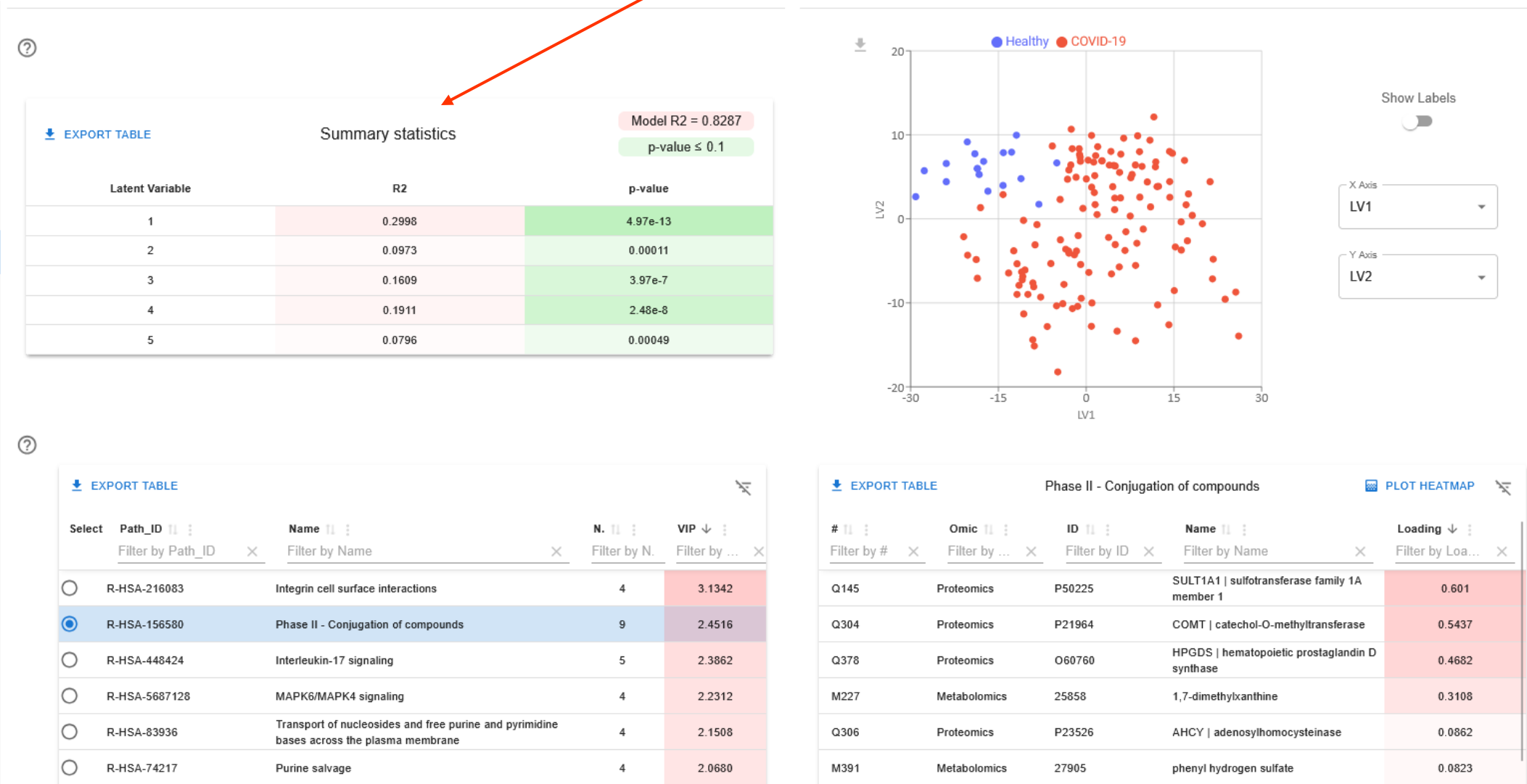
To the right of the main steps is an 'Optional' section titled 'Complement the analysis with your Pathways (Optional)'. It includes an 'UPLOAD NEW PATHWAY' button, a text field showing 'Uploaded: BA_mouse.gmt', and a 'Select Pathways' dropdown menu with 'BA_human, BA_mouse' selected. Below the dropdown are 'BA_human' and 'BA_mouse' options. An annotation box below this section says: 'Users can upload **custom pathways in GMT format**, enabling the integration of personalized metabolite-based pathway definitions'.

Single-View approach computes more comprehensive pathway scores, using Partial Least Squares Discriminant Analysis (PLS-DA) on a single pathway-level matrix that combines information from all omics

Pathway Integrative Analysis

➤ Single-View Framework results

R-squared (R^2) value representing the proportion of variance in the response variable explained by the model. An **empirical p-value** is, calculated by performing ten permutations on the response vector (y) and comparing the simulated R^2 values with the actual one. The minimum p-value that can be obtained is 0.1, reflecting the reliability of the model.



Pathway Integrative Analysis

➤ Single-View Framework results

DATA DISTRIBUTION

PCA

MULTIOMICS FACTOR ANALYSIS

ENRICHMENT ANALYSIS

PATHWAY INTEGRATIVE ANALYSIS

a)

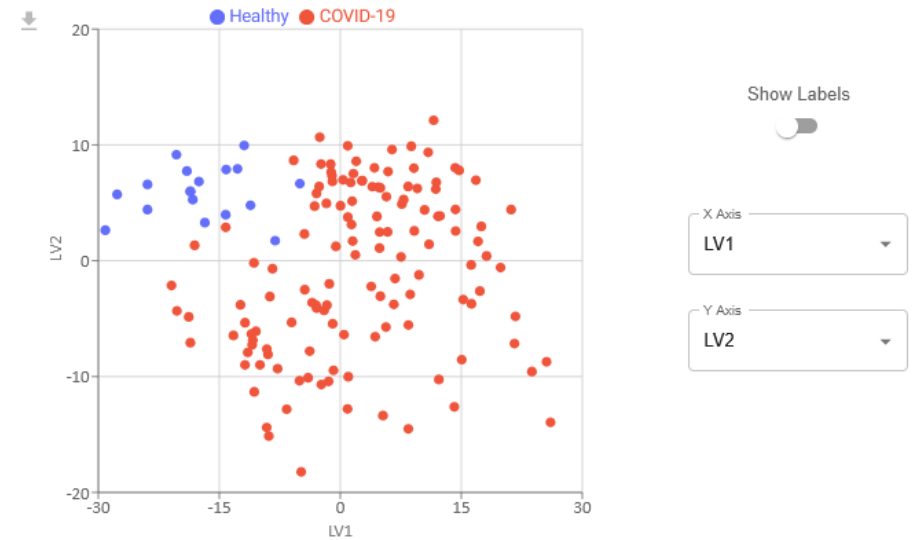
EXPORT TABLE

Summary statistics

Model R2 = 0.8287

p-value ≤ 0.1

Latent Variable	R2	p-value
1	0.2998	4.97e-13
2	0.0973	0.00011
3	0.1609	3.97e-7
4	0.1911	2.48e-8
5	0.0796	0.00049



Click on the “**EXPORT TABLE**” to download the results tables:

a) Pathways analysis model with statistical summary

b) Relevant Pathway displaying the pathways most strongly associated with the outcome;

c) Pathways exploration, with biomolecules associated with a specific pathway

EXPORT TABLE b)

Select	Path_ID	Name	N.	VIP
	Filter by Path_ID	Filter by Name	Filter by N.	Filter by ...
☐	R-HSA-216083	Integrin cell surface interactions	4	3.1342
☒	R-HSA-156580	Phase II - Conjugation of compounds	9	2.4516
☐	R-HSA-448424	Interleukin-17 signaling	5	2.3862
☐	R-HSA-5687128	MAPK6/MAPK4 signaling	4	2.2312
☐	R-HSA-83936	Transport of nucleosides and free purine and pyrimidine bases across the plasma membrane	4	2.1508
☐	R-HSA-74217	Purine salvage	4	2.0680

EXPORT TABLE c)

Phase II - Conjugation of compounds

PLOT HEATMAP

#	Omic	ID	Name	Loading
Filter by #	Filter by ...	Filter by ID	Filter by Name	Filter by Loa...
Q145	Proteomics	P50225	SULT1A1 sulfotransferase family 1A member 1	0.601
Q304	Proteomics	P21964	COMT catechol-O-methyltransferase	0.5437
Q378	Proteomics	O60760	HPGDS hemopoietic prostaglandin D synthase	0.4682
M227	Metabolomics	25858	1,7-dimethylxanthine	0.3108
Q306	Proteomics	P23526	AHCY adenosylhomocysteinase	0.0862
M391	Metabolomics	27905	phenyl hydrogen sulfate	0.0823

Pathway Integrative Analysis

➤ Multi-View Framework

The screenshot shows the 'PATHWAY INTEGRATIVE ANALYSIS' section of a web application. The interface includes a navigation bar with links like 'Home', 'TurboPutative', 'TurbOmics', 'Web Services', 'Help', and 'Contact us'. A search bar is in the top right. The main content area has a sidebar with options: 'Main page', 'DATA DISTRIBUTION', 'PCA', 'MULTIOMICS FACTOR ANALYSIS', 'ENRICHMENT ANALYSIS', and 'PATHWAY INTEGRATIVE ANALYSIS'. The main panel has tabs for 'Single-View' and 'Multi-View'. Below these are three numbered steps: (1) '1. Select the Metadata' with a 'Group' dropdown and two 'Group' boxes (Group 1: 'He...', Group 2: 'C...'); (2) '2. Select the Omics Mapping' with 'Metabolomics ID Column' (ChEBI_ID) and 'Proteomics ID Column' (UniProt_ID) dropdowns, and corresponding 'ID Type' dropdowns; (3) 'RUN ANALYSIS >' button. An 'Optional' section on the right allows uploading a new pathway (BA_mouse.gmt) and selecting pathways (BA_human, BA_mouse). Annotations with dashed boxes and arrows point to specific elements: an orange box around step (1) with the text 'Specify the comparison groups to be considered in the pathways analysis'; a blue box around step (2) with the text 'Select the Identifiers for each omic layer to use in the pathways analysis'; and a green box around the optional section with the text 'Users can upload custom pathways in GMT format, enabling the integration of personalized metabolite-based pathway definitions'.

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Search Job

Job ID: 72f42rH8UT

Main page

DATA DISTRIBUTION

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MULTIOMICS FACTOR ANALYSIS

ENRICHMENT ANALYSIS

PATHWAY INTEGRATIVE ANALYSIS

Single-View Multi-View

(1) 1. Select the Metadata

Metadata Column

Group

Group 1 Group 2

He... C...

(2) 2. Select the Omics Mapping

Metabolomics ID Column Proteomics ID Column

ChEBI_ID UniProt_ID

ChEBI Uniprot Protein ID

ID Type ID Type

(3) RUN ANALYSIS >

Optional

Complement the analysis with your Pathways (Optional)

UPLOAD NEW PATHWAY

Uploaded: BA_mouse.gmt

Select Pathways

BA_human, BA_mouse

Choose your custom pathways

Specify the **comparison groups** to be considered in the pathways analysis

Select the **Identifiers** for each omic layer to use in the pathways analysis

Users can upload **custom pathways in GMT format**, enabling the integration of personalized metabolite-based pathway definitions

Multi-View approach employs a multi-block partial least squares (MB-PLS) latent variable model. It preserves the block structure of each omics view, enabling you to see the contribution of each omics dataset to the prediction of the outcome variable.

Pathway Integrative Analysis

➤ Multi-View Framework results

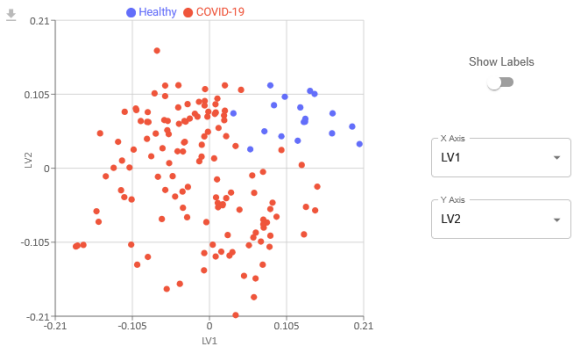
a)

EXPORT TABLE

Summary statistics

Model R2 = 0.7864
p-value ≤ 0.1

Latent Variable	Metabolomics (%)	Proteomics (%)	R2	p-value
1	16.44	83.56	0.3007	4.54e-13
2	55.46	44.54	0.1025	6.88e-5
3	53.56	46.44	0.142	2.17e-6
4	85.37	14.63	0.1777	8.60e-8
5	88.31	11.69	0.0635	0.00194



DATA DISTRIBUTION

PCA

MULTIOMICS FACTOR ANALYSIS

ENRICHMENT ANALYSIS

PATHWAY INTEGRATIVE ANALYSIS

Click on the **“EXPORT TABLE”** to download results :
a) Pathways analysis model with statistical summary;
b) Relevant Pathway displaying the pathways most strongly associated with the outcome;
c) Pathways exploration with biomolecules associated with a specific pathway

?

EXPORT TABLE

b1) Metabolomics

Select	Path_ID	Name	N.	VIP
☐	R-HSA-373076	Class A/1 (Rhodopsin-like receptors)	6	2.5551
☒	R-HSA-156580	Phase II - Conjugation of compounds	5	2.5417
☐	R-HSA-556833	Metabolism of lipids	11	2.5026
☐	R-HSA-416476	G alpha (q) signalling events	4	2.4952
☐	R-HSA-1483206	Glycerophospholipid biosynthesis	4	2.4916
☐	R-HSA-425393	Transport of inorganic cations/anions and amino acids/oligopeptides	5	2.4465
☐	R-HSA-5619102	SLC transporter disorders	11	2.4452

EXPORT TABLE

c1) Phase II - Conjugation of compounds

PLOT HEATMAP

#	ID	Name	Loading
M227	25858	1,7-dimethylxanthine	0.7268
M391	27905	phenyl hydrogen sulfate	0.3355
M710	37946	dopamine 3-O-sulfate	-0.065
M82	9008	salicylic acid	-0.0822
M435	17864	taurothiocholic acid sulfate	-0.5902

EXPORT TABLE

b2) Proteomics

Select	Path_ID	Name	N.	VIP
☐	R-HSA-3007120	MAPK9/MAPK4 signaling	4	1.5404
☐	R-HSA-5669034	TNFs bind their physiological receptors	7	1.5163
☒	R-HSA-156580	Phase II - Conjugation of compounds	4	1.5101
☐	R-HSA-8957322	Metabolism of steroids	4	1.5058
☐	R-HSA-8979227	Triglyceride metabolism	4	1.4659
☐	R-HSA-163560	Triglyceride catabolism	4	1.4659
☐	R-HSA-168255	Influenza Infection	4	1.4357
☐	R-HSA-5218859	Regulated Necrosis	8	1.4325

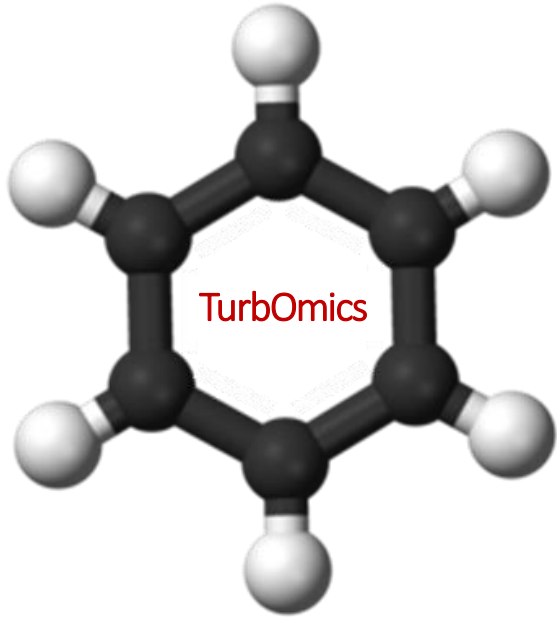
EXPORT TABLE

c2) Phase II - Conjugation of compounds

PLOT HEATMAP

#	ID	Name	Loading
Q145	P50225	SULT1A1 sulfotransferase family 1A member 1	0.6379
Q304	P21964	COMT catechol-O-methyltransferase	0.6269
Q378	O60760	HPGDS hematopoietic prostaglandin D synthase	0.3758
Q306	P23526	AHCY adenosylhomocysteinase	0.2426

Summary



For additional details and explanation, please visit:
<https://proteomics.cnice.es/TurboPutative/turbomicshelp>

If you have any questions, please contact us:
metabolomics.cnice@gmail.com

- TurbOmics is an intuitive web-based platform designed to facilitate the **multi-omics analysis of metabolomic/lipidomics data alongside proteomics and transcriptomics**.
- It supports both **targeted** experiments and those where metabolite **identification** is available (e.g., LC/CE-MS/MS, NMR or GC-MS data).
- Additionally, TurbOmics offers tools to **putatively annotate features** from untargeted metabolomics based on their mass-to-charge (m/z) ratio.
- The platform provides a streamlined workflow for **exploratory data analysis** and **multi-omics integration** through i) **Multi-Omics Factor Analysis**, ii) **Pathway Integrative Analysis** and iii) **Enrichment Analysis**.
- Users can upload their experimental data directly from the **Main Page**. Previous analyses can be accessed using the **Find Job** field by entering the corresponding job ID.
- Once a dataset is loaded, the **Results** section provides access to a suite of tools for in-depth analysis and interpretation where users can explore and export data tables and visualize plots of the results.